DSPi enclosure: printing and assembly

A 3D-printed enclosure for the USB DSP sound card built from a Raspberry Pi Pico 2 H and a GY-PCM5102 module – printing, gluing the modules, wiring, lid and sticker



Fig. 1 – The finished enclosure with the light laminated sticker on the lid.

What you build	a closed 70.8 × 68.8 × 34 mm box powered and controlled by a single USB cable; micro-USB at the back, 3.5 mm output jack on the right side
Printed parts	base and lid (2 STL files), no supports; roughly 40–50 g of PLA or PETG
Module mounting	the Pico and the DAC drop into printed cradles and are glued – no M2 screws needed
Assembly time	about 30–45 minutes including soldering the header onto the DAC module
Tools	soldering iron, Phillips screwdriver, hot-glue gun or silicone / plastic glue, knife or deburring tool
Files	repository github.com/hanezzzz00/dspi-pico2-diy: *enclosure/stl* (STL), *enclosure/fusion* (Fusion script), *sticker* (sticker PDF)
Version	enclosure v3 (25 Oct 2026): DAC at the right wall, glue cradle for the Pico instead of screws; companion to the guide "DSPi: a digital sound processor for the price of a pizza or two"

1. What you need

The electronics are the same as in the main guide. The enclosure is designed for the **Raspberry Pi Pico 2 H** (with soldered headers) and the common purple **GY-PCM5102** module of roughly 32 × 17 mm, whose jack sticks out of the **long edge** of the board at the end opposite the 6-pin header.



Fig. 2 – Raspberry Pi Pico 2 H (≈ €7).



Fig. 3 – GY-PCM5102 module (≈ €4); jack on the long edge.

Item	Qty	Price	Note
Raspberry Pi Pico 2 H	1	≈ €7	headers soldered at the factory
GY-PCM5102 DAC module (≈ 32 × 17 mm)	1	≈ €4–6	6-pin header included, you solder it yourself
M3 × 10–12 countersunk screw	4	≈ €1*	lid; self-tapping into the plastic (Ø2.5 hole)
Hot glue, silicone or plastic glue	—	—	fixing the Pico and DAC in their cradles
Female–female jumper wires, 10 cm	5	≈ €1*	shorter wires are easier to tidy inside
Micro-USB data cable	1	≈ €2–6*	plug up to 12 × 8 mm
3.5 mm jack – 2× RCA or jack–jack cable	1	≈ €2–4*	plug up to Ø10 mm
PLA or PETG filament	≈ 50 g	≈ €1*	any colour
Optional: self-adhesive rubber feet	4	≈ €1*	keeps the box from sliding
Optional: paper for the sticker + laminating pouch	1	—	ch. 7

* indicative price. Shops are intentionally not listed; the same parts are sold by many vendors.

Different DAC module?
PCM5102A modules from different vendors differ in size and jack position. Measure yours; if it differs, adjust the parameters per chapter 3 before printing. A board of 17.35 × 31.8 mm still fits the cradle.

2. How the enclosure is built

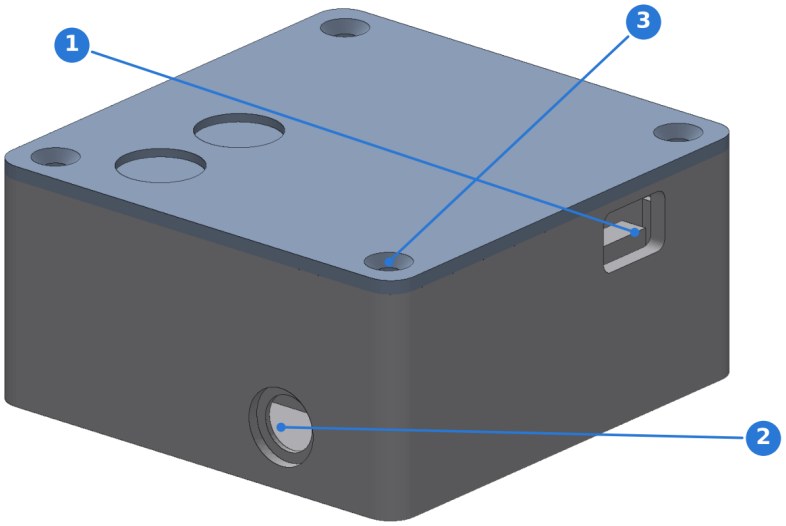


Fig. 4 – Assembled enclosure: (1) micro-USB opening in the back wall, (2) 3.5 mm jack in the right wall, (3) countersunk lid screws. The lid has recesses for two future buttons.

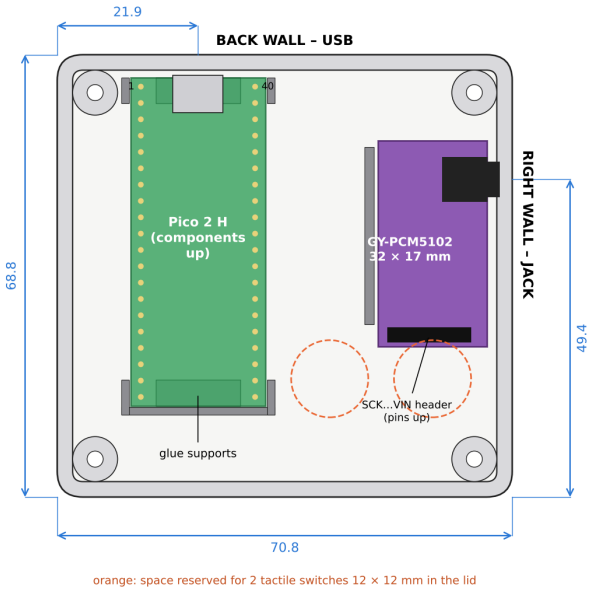


Fig. 5 – Top view layout, back wall at the top.

- **Pico 2 H** lies components up in a cradle 22 mm above the floor. The supports sit under both ends of the board **between the pin rows**, so neither the pins nor the jumper sockets hit anything. A front stop and side stops locate the board; you just glue it.
- The Pico's **micro-USB** faces a 12.5 × 8.5 mm opening in the back wall.
- The **DAC module** lies at the right wall on supports 2.5 mm above the floor; the jack points out of the right side through a Ø8 mm hole with a Ø11 mm counterbore outside. The header (SCK … VIN) is at the front end, pins up.
- The **lid** is held by four M3 screws in the corner posts. Underneath it has frames for two tactile switches; there is room for them in front of the modules (orange).

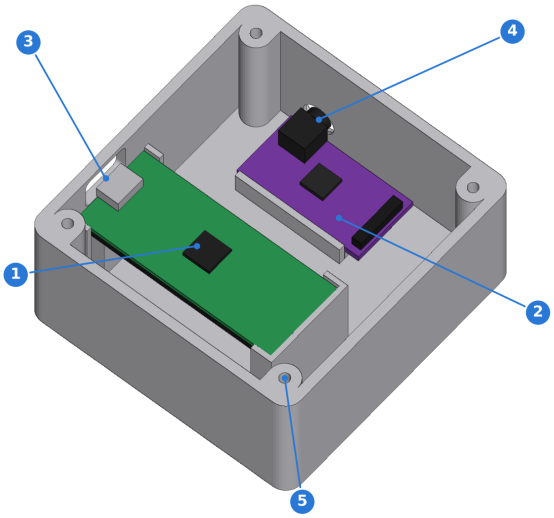


Fig. 6 – Base with modules: (1) Pico, (2) DAC, (3) USB, (4) jack, (5) corner post.

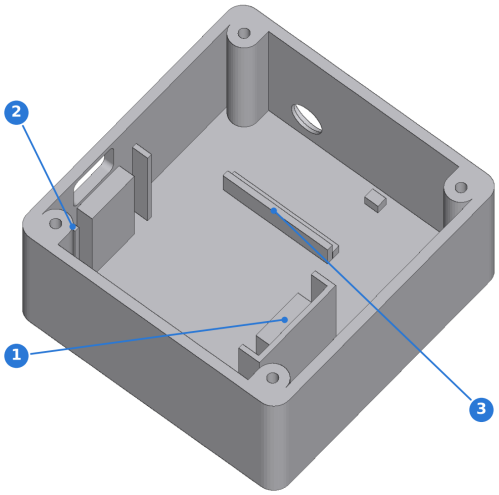


Fig. 7 – Empty base: (1) Pico cradle support, (2) side stop, (3) DAC supports.

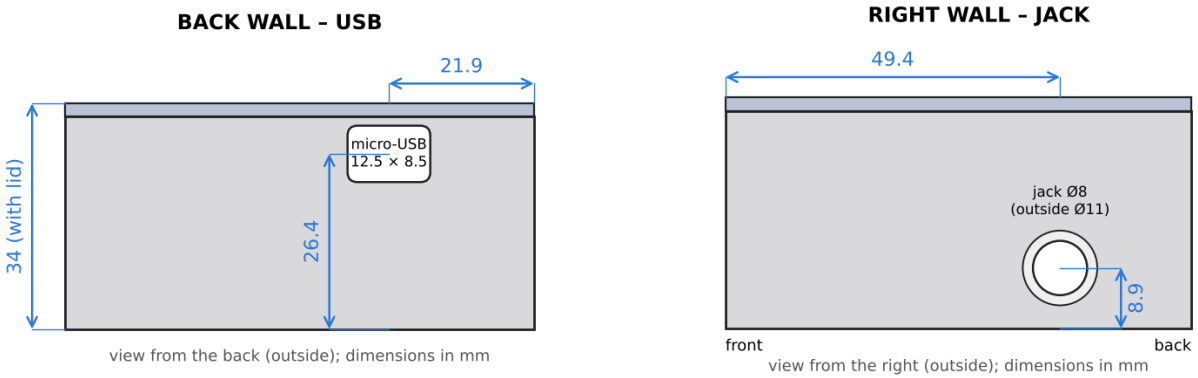


Fig. 8 – Hole positions from outside (mm): left the back wall with USB, right the right wall with the jack.

Parameter	Value
Outer size	70.8 × 68.8 × 34 mm (base 32 mm + lid 2 mm)
Inner space	66 × 64 × 30 mm; walls 2.4 mm, floor and lid 2 mm, vertical edges R4
USB opening	12.5 × 8.5 mm, R1.5; centre 26.4 mm above the bottom, 21.9 mm from the left back edge
Jack hole	Ø8 mm, Ø11 mm counterbore outside (0.8 mm wall left); centre 8.9 mm above the bottom, 49.4 mm from the front edge
Pico cradle	underside of the board 22 mm above the floor; 0.25 mm clearance per side; 1.2 mm stops
Lid screws	4× M3 (Ø2.5 hole, 12 mm deep; Ø3.4 in the lid with Ø6.5 countersink)
Button provision	2× Ø12 mm recess in the lid, 0.8 mm skin left; 12.3 mm frame; M2 bosses ±11 mm

3. Modifying the model (only if needed)

The model is generated by the *DSPi_krabicka.py* script for Autodesk Fusion. At the top of the script is a **PARAMETRY** (parameters) section with all dimensions in millimetres. If the defaults suit you, print the supplied STL files and skip this chapter.

Parameter	Default	Meaning
DAC_W, DAC_L	17 / 32	width and length of the DAC board
JACK_FROM_END	6	jack centre from the back end of the DAC board (measure)
JACK_Z_ABOVE_PCB	2.8	jack centre height above the top of the board (measure)
DAC_Y0, DAC_Z	21 / 2.5	position of the DAC front edge and support height
JACK_HOLE_D, JACK_CBORE_D	8 / 11	jack hole and counterbore for the cable plug
PICO_POST_H	22	height of the Pico underside above the floor (room for pins and sockets)
PICO_FIT	0.25	clearance of the Pico in the cradle per side
USB_OPEN_W, USB_OPEN_H	12.5 / 8.5	micro-USB opening; enlarge for a bulky plug
CPOST_PILOT	2.5	lid screw hole; for M3 heat-set inserts set it per the insert (usually 4.0)

Running it in Fusion

1. In Fusion open **Utilities → Add-Ins → Scripts and Add-Ins** (Shift+S).
2. Click **+** → *Script or add-in from device* and select the *DSPi_krabicka* folder.
3. Next to **DSPi_krabicka** click **Run**. The script creates a new design with base and lid components and saves both STL files into the same folder.
4. After editing parameters in a text editor, simply run the script again.

4. Printing

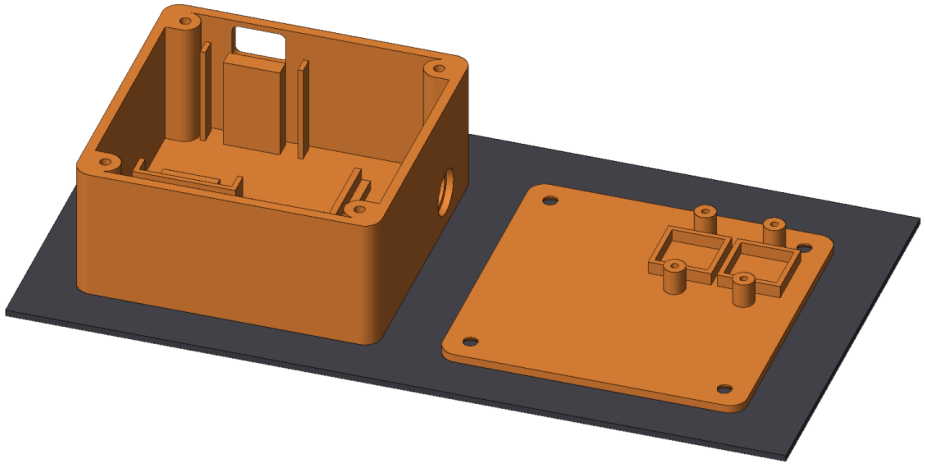


Fig. 9 – Plate layout: base floor down, lid outer face down. No supports needed.

Flip the lid
The lid is stored in the STL in its assembled position, outer face up. In the slicer lay it outer face down on the plate (in PrusaSlicer the **Place on face** tool, key F), or rotate it 180° around the X axis.

Setting	Recommendation	Why
Material	PLA or PETG	PETG copes better with heat, e.g. behind a computer
Nozzle / layer	0.4 / 0.2 mm	standard settings
Perimeters	3 (better 4)	strong posts for the M3 screws
Infill	20 % (gyroid)	the box is not loaded
Top / bottom layers	5 / 4	smooth floor and lid
Supports	off	the USB bridge is only 12.5 mm, round holes print fine
Elephant-foot compensation	on (0.1–0.2 mm)	the lid seats better on the base

After printing

- Remove strings from the openings and trim any first-layer flare around the perimeter.
- Drive the M3 screws in and out once without electronics – this cuts the thread in the plastic.
- Dry-fit: the Pico must drop into the cradle between the stops, the DAC onto its supports and the jack plug into the hole fully.

5. Preparing the electronics

> **Test first, glue later**
> Before gluing anything, flash the DSPi firmware, wire everything up on the desk and test the sound per the main guide (ch. 6–8). Mistakes are much easier to find on the desk.

DAC header: pins up

The module rests on supports, so pins pointing down would hit the floor. Insert the header **from the top side of the board** (pins pointing up) and solder from below. Alternatively solder the wires directly into the holes. Also check the solder bridges on the back (H1L–H4L, SCK) per the main guide, ch. 6.

Wiring

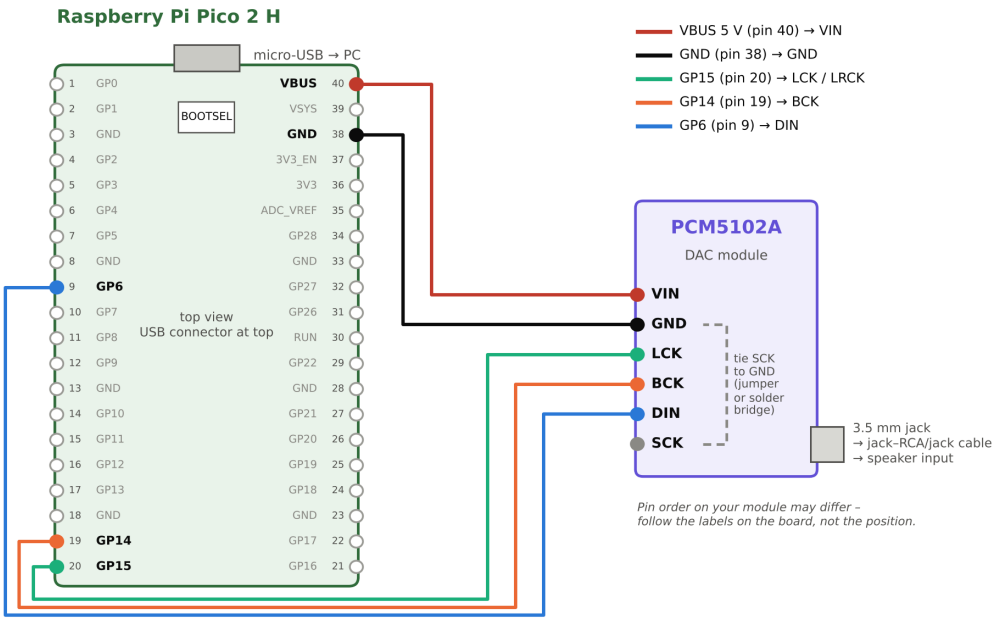


Fig. 10 – Wiring the Pico 2 H and the PCM5102A module (from the main guide).

Wire	Pico 2 H (pin / function)	GY-PCM5102	Purpose
red	pin 40 / VBUS (5 V)	VIN	power
black	pin 38 / GND	GND	ground
orange	pin 19 / GP14	BCK	bit clock
green	pin 20 / GP15	LCK	channel clock
blue	pin 9 / GP6	DIN	data

6. Step-by-step assembly

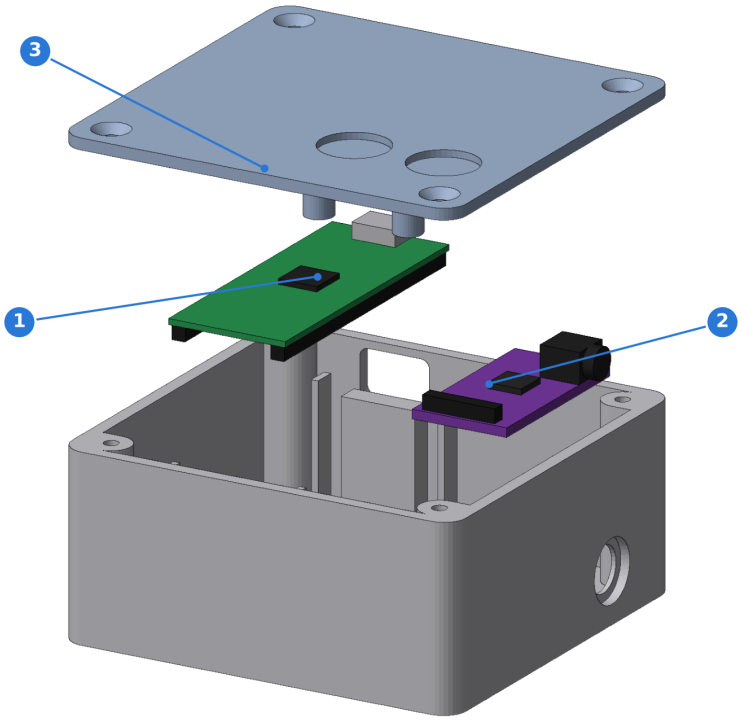


Fig. 11 – Assembly order: (1) Pico into the cradle, (2) DAC at the right wall, (3) lid.

1. **Attach the wires.** Push the sockets onto Pico pins 40, 38, 19, 20 and 9 and onto the DAC pins while still outside the box.
2. **Insert the DAC.** Place it on the supports at the right wall with the jack facing the hole and plug in the jack cable – it must go all the way. Leave the cable plugged in; it holds the module in position.
3. **Glue the DAC.** A drop of hot glue or silicone on the supports near the front end of the board. Keep glue out of the jack and off the header.
4. **Insert the Pico.** Components up, micro-USB towards the back-wall opening, into the cradle between the stops. Plug in the USB cable and check that it goes all the way.
5. **Glue the Pico.** A small drop on each support under the ends of the board, or where the board meets a stop. Do not glue over the pins. The BOOTSEL button will no longer be reachable, but firmware can be updated from DSPi Console (☰ → Update Firmware) without it.
6. **Test the sound.** Connect USB, play music and wiggle the cables – the sound must not drop out.
7. **Tuck the wires** along the side wall so they do not obstruct the lid.
8. **Close the lid** with the button frames towards the front right (over the free space in front of the modules) and drive the four M3 screws crosswise, gently.
9. **Finish.** Stick on the rubber feet and, optionally, the sticker (ch. 7).

> **Which glue**
> Hot glue can be released with a drop of isopropyl alcohol or careful warming with a hair dryer. Silicone holds more flexibly and can be cut with a knife. Avoid superglue – its fumes leave a white haze.

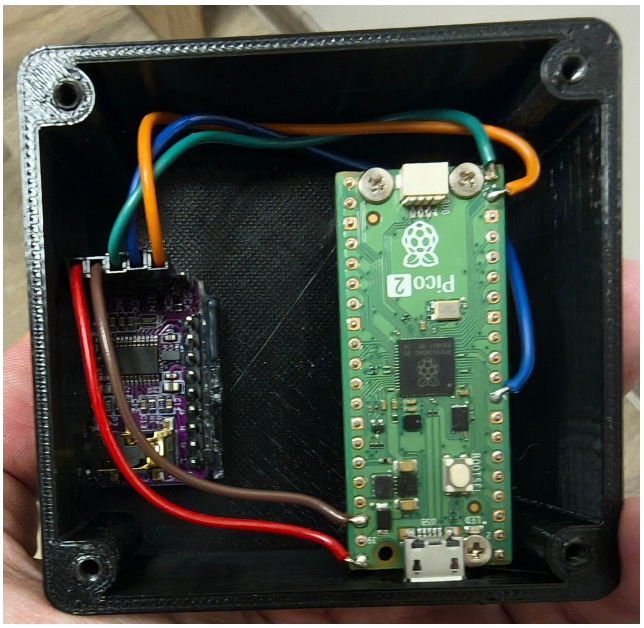


Fig. 12 – Prototype from above (Pico still on M2 screws, version v2).

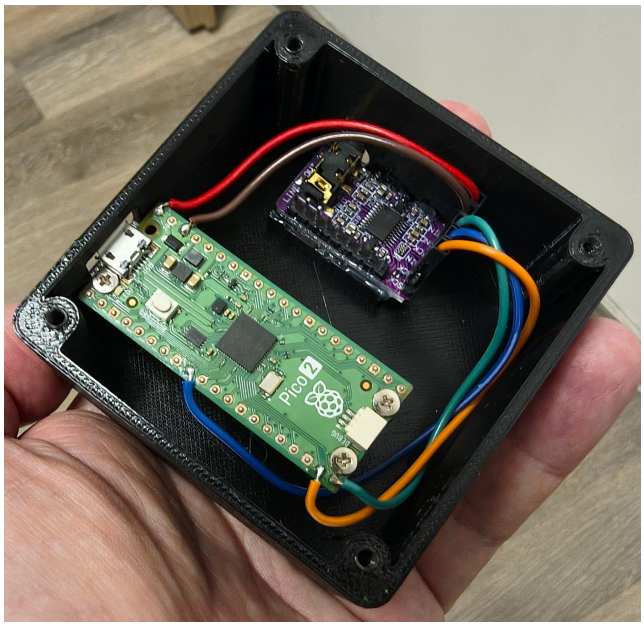


Fig. 13 – Prototype: DAC at the right wall, wires tucked to the sides.

The photos show a prototype in which the Pico was screwed down with M2 screws. The current v3 replaces the posts with a cradle whose supports sit between the pin rows, and the Pico is glued; layout and wiring are the same.

7. Lid sticker

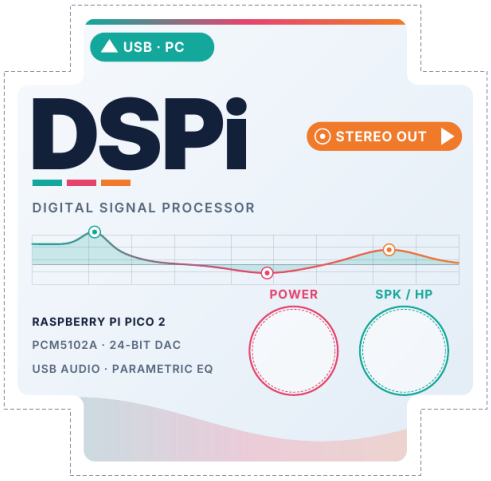


Fig. 14 – Light full-lid sticker (corner notches for the screws).

- The *sticker* folder has two versions: **light, covering the whole lid** (recommended) and **dark** with smaller labels and button strips.
- Print at **100 %** scale (disable "fit to page"). Use the 50 mm scale bar on the page to check.
- The light version has a cut outline and artwork inset by 2 mm – the margin is for the laminating film overlap. Laminate, cut along the outline with scissors and cut out the corner notches for the screws.
- Attach with double-sided tape or spray adhesive. The POWER and SPK / HP rings sit over the button recesses, the USB and STEREO OUT labels point to the connectors.
- Without laminating, a few thin coats of clear acrylic spray varnish will do.



Fig. 15 – The printed and laminated light sticker on the lid.

8. Preparing for buttons (when you want them)

The lid has two places for 12 × 12 mm tactile switches: e.g. button 1 for mute or an "Off" preset and button 2 for switching speakers / headphones. Until you need them, the recesses stay closed by a 0.8 mm skin.

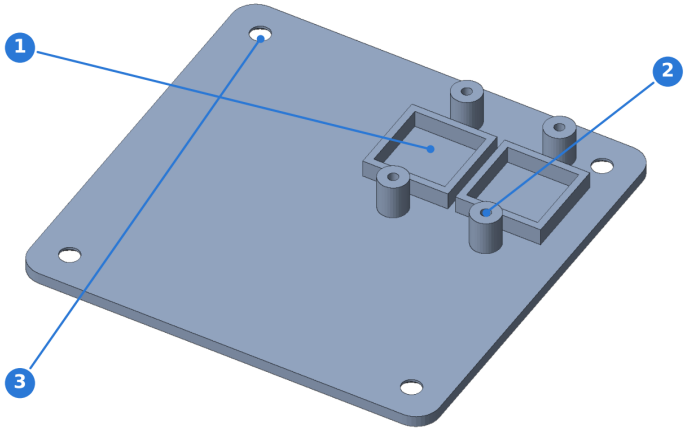


Fig. 16 – Underside of the lid: (1) switch frame, (2) M2 bosses for the switch board, (3) screw hole.

- **Hole:** drill through the skin with a Ø12 mm or step drill (through the sticker if fitted).
- **Mounting:** the switch soldered to a small piece of perfboard slides into the frame from below and the board is screwed to the two bosses.
- **Wiring:** one switch lead to a free Pico pin (e.g. GP18 = pin 24, GP19 = pin 25), the other to GND.
- **Setup:** in DSPi Console in the *Control Surfaces* window (main guide, ch. 10.4).

9. Troubleshooting

Problem	Likely cause	Fix
Micro-USB will not plug in	bulky plug; Pico shifted in the cradle	push the Pico against the back wall; file the opening or enlarge USB_OPEN_W / H
Jack will not go all the way	cable plug wider than Ø11 mm	slimmer cable; or enlarge JACK_CBORE_D
Jack not centred in the hole	the module's jack is elsewhere or at a different height	measure and adjust JACK_FROM_END and JACK_Z_ABOVE_PCB
Pico too tight or loose in the cradle	printer tolerance	adjust PICO_FIT (0.15–0.4)
Lid screw spins	stripped thread	longer screw or heat-set insert (CPOST_PILOT per insert)
No sound after closing	a wire slipped off a pin	open and check all 5 wires
Lid does not seat	lid rotated or first-layer flare	rotate the lid (ch. 6, step 8); chamfer the base edge

License and links

The enclosure, sticker and this guide are licensed under CC BY-SA 4.0. The DSPi firmware is the work of Weeb Labs under GPL-3.0. Latest files: https://github.com/hanezzzz00/dspi-pico2-diy

- DSPi – firmware and documentation: https://github.com/WeebLabs/DSPi
- Raspberry Pi Pico 2 pinout (PDF): https://datasheets.raspberrypi.com/pico/Pico-2-Pinout.pdf